

adherent cells, amplified and retransfected. Four consecutive cycles of library selection were completed.

Cytotoxic treatments, kinase assays and antibodies

Plasmids were transiently transfected into RelA^{-/-} cells, and after 24 h cultures were treated with CHX (0.1 µg ml⁻¹) with or without TNF-α (1,000 U ml⁻¹; Peprotech). In Fig. 4d, plasmids were co-transfected with pEGFP (Clontech). At the indicated times, we counted and analysed adherent cells by flow cytometry (FCM) to determine the numbers of live GFP-positive cells. With 3DO clones, apoptosis was assessed by staining with propidium iodide (Sigma) followed by FCM analysis, as described³⁰. Unless otherwise stated, 3DO cells were treated with 25 U ml⁻¹ TNF-α. In Fig. 4e, 3DO-IκBαM cells were pretreated with mitogen-activated protein kinase (MAPK) inhibitors (Calbiochem) for 30 min and then incubated with either TNF-α (25 U ml⁻¹) or PBS for an additional 12 h. In both RelA^{-/-} MEFs and 3DO cells, percentages of live cells in experimental cultures were calculated relative to the number of live cells present in mock-treated cultures. We performed kinase assays with recombinant glutathione S-transferase (GST)-c-Jun and anti-JNK antibodies (Pharmingen), as described²³. In all cases, cells were treated with 1,000 U ml⁻¹ TNF-α. In Fig. 4c, CHX treatments (10 µg ml⁻¹) were carried out for 30 min in addition to the indicated time. We used the following antibodies: phospho-JNK (Fig. 4b), IκBαM and β-actin from Santa Cruz Biotechnology; Bid, caspase-6, -7 and -9, and total and phospho-JNK, -ERK and -p38 from Cell Signaling Technology; caspase-8 from Alexis; caspase-2 and -3 from R&D Systems. The anti-Gadd45β antibody was generated against an N-terminal peptide of murine Gadd45β.

Caspase activity and transmembrane potential

We performed *in vitro* fluorimetric assays of caspase activity as described³⁰, by incubating cell lysates with amino trifluoromethyl coumarin (ATC)-labelled caspase-specific peptides (Bachem): zVDVAD (caspase-2), zDEVD (caspases-3/7), zVEID (caspase-6), zETD (caspase-8) and Ac-LEHD (caspase-9). The mitochondrial transmembrane potential was measured by using the fluorescent dye JC-1 (Molecular Probes) and FCM, according to the manufacturer's instructions.

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Inhibition of JNK activation through NF-κB target genes

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The proinflammatory cytokine tumour necrosis factor-α (TNF-α) regulates immune responses, inflammation and programmed cell death (apoptosis)^{1–4}. The ultimate fate of a cell exposed to TNF-α is determined by signal integration between its different effectors, including IκB kinase (IKK), c-Jun N-terminal protein kinase (JNK) and caspases¹. Activation of caspases is required for apoptotic cell death⁵, whereas IKK activation inhibits apoptosis through the transcription factor NF-κB, whose target genes include caspase inhibitors^{1,6–10}. JNK activates the transcription factor c-Jun/AP-1, as well as other targets^{11–16}. However, the role of JNK activation in apoptosis induced by TNF-α is less clear^{17,18}. It is unknown whether any crosstalk occurs between IKK and JNK, and, if so, how it affects TNF-α-induced apoptosis. We investigated this using murine embryonic fibroblasts that are deficient in either the IKKβ catalytic subunit of the IKK complex or the RelA/p65 subunit of NF-κB. Here we show that in addition to inhibiting caspases, the IKK/NF-κB pathway negatively modulates TNF-α-mediated JNK activation, partly through NF-κB-induced X-chromosome-linked inhibitor of apoptosis (XIAP)^{7,9}. This negative crosstalk, which is specific to TNF-α signalling and does not affect JNK activation by interleukin-1 (IL-1), contributes to inhibition of apoptosis.

IKKβ is required for activation of NF-κB by TNF-α, whereas

IKK α is dispensable^{6,19,20}. To determine whether interfering with the IKK/NF- κ B signalling pathway affects JNK activation by TNF- α , we used wild-type (WT), IKK α -deficient (*Ikk α ^{-/-}*)¹⁹ and IKK β -deficient (*Ikk β ^{-/-}*)²⁰ mouse embryonic fibroblasts (Fig. 1a). After treatment with TNF- α , the activities of IKK and JNK were measured by immune-complex kinase assays^{15,21}. Activation of IKK by TNF- α was transient and robust in WT fibroblasts and *Ikk α ^{-/-}* cells, but severely impaired in *Ikk β ^{-/-}* cells (Fig. 1b). By contrast, JNK activation was transient in WT and *Ikk α ^{-/-}* fibroblasts but long lasting in *Ikk β ^{-/-}* cells (Fig. 1c). This sustained activation did not result from increased expression of either p46^{JNK} (Fig. 1c) or p54^{JNK} (data not shown). Transfection of *Ikk β ^{-/-}* cells with an IKK β expression vector restored transient JNK activation in response to TNF- α (Fig. 1d). Loss of IKK β did not affect TNF- α -mediated activation of p38 MAPK (Fig. 1e) or ERK (Fig. 1f).

To investigate the role of NF- κ B in IKK-mediated JNK inhibition, we used RelA-deficient (*RelA^{-/-}*) mouse embryonic fibroblasts (Fig. 2a) lacking the major activating subunit of NF- κ B²². Immune-complex kinase assays showed that TNF- α induced sustained JNK activation in *RelA^{-/-}* fibroblasts (Fig. 2b). Even basal JNK activity was slightly higher in *RelA^{-/-}* cells than in WT controls (Fig. 2b). The sustained activation was not the result of increased JNK expression (Fig. 2b). Transfection of *RelA^{-/-}* fibroblasts with a RelA expression vector restored transient JNK activation in response to TNF- α stimulation (Fig. 2c).

Because JNK is a key regulator of c-Jun activity¹¹, its negative regulation by the IKK/NF- κ B pathway might affect c-Jun-mediated transcription. Indeed, TNF- α stimulated the transcriptional activity of a GAL4-c-Jun chimera, in which the GAL4 DNA binding domain was fused to the c-Jun transactivation domain¹⁵, by 20-

fold in *RelA^{-/-}* cells but only 7-fold in WT controls (Fig. 2d). Transfection of *RelA^{-/-}* fibroblasts with a RelA expression vector, which activated an NF- κ B reporter gene (Fig. 2d), abrogated the stimulation of GAL4-c-Jun activity by TNF- α (Fig. 2d). This suggests that the augmented c-Jun activation is due to the loss of NF- κ B-mediated JNK inhibition.

To determine whether gene expression induced by NF- κ B is required to block JNK activation, WT fibroblasts were treated with TNF- α and the protein synthesis inhibitor cycloheximide (CHX) or CHX alone. In the presence of CHX, TNF- α induced prolonged JNK activation (Fig. 2e). The prolonged activation of JNK by TNF- α in CHX-treated WT fibroblasts was less sustained than in *RelA^{-/-}* cells (Fig. 2b). This was probably due to incomplete inhibition of *de novo* protein synthesis by CHX at the concentration (10 μ g ml⁻¹) used (data not shown). The sustained activation of JNK by TNF- α plus CHX did not result from increased expression of JNK, as examined by immunoblotting (data not shown). Although CHX can activate JNK in some cell types, it had a marginal effect on JNK activity in mouse fibroblasts (Fig. 2e). It seems that NF- κ B-mediated suppression of JNK activation requires *de novo* synthesis of JNK inhibitors.

Dephosphorylation is a probable mechanism for inactivation of JNK^{23,24}. We examined whether NF- κ B inhibits JNK activation through induction of JNK phosphatases. Incubation of purified active JNK with extracts from either non-treated WT or *RelA^{-/-}* cells resulted in partial JNK inactivation (Fig. 3a, ~50% reduction). Treatment of either WT or *RelA^{-/-}* cells with TNF- α did not increase JNK phosphatase activity (Fig. 3a). These results, however, do not exclude the possibility that NF- κ B may induce phosphatases that inactivate upstream JNK activators. We therefore examined JNK

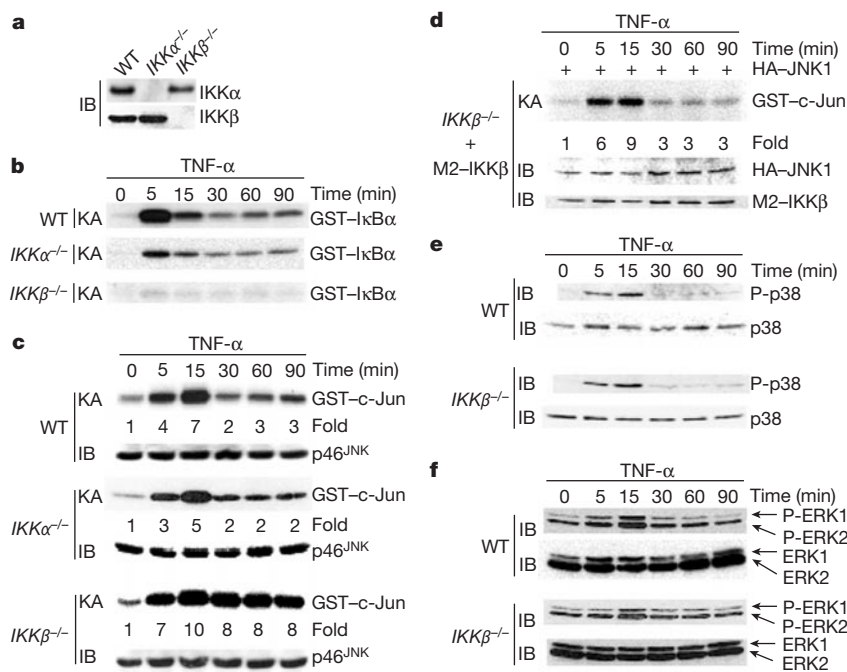


Figure 1 Genetic disruption of IKK β results in sustained JNK activation by TNF- α . **a**, Immunoblot (IB) analysis of IKK α and IKK β expression in WT, *Ikk α ^{-/-}* and *Ikk β ^{-/-}* fibroblasts with anti-IKK α (Santa Cruz) and anti-IKK β (Upstate) antibodies, respectively. **b**, Activation of IKK by TNF- α is impaired in *Ikk β ^{-/-}* fibroblasts. WT, *Ikk α ^{-/-}* and *Ikk β ^{-/-}* fibroblasts were treated with TNF- α (20 ng ml⁻¹) as indicated. IKK was immunoprecipitated with anti-IKK γ antibody (Santa Cruz) and its activity measured by immune-complex kinase assay (KA) using glutathione *S*-transferase (GST)-I κ B α (1–54) as substrate²¹. **c**, Activation of JNK by TNF- α is prolonged in *Ikk β ^{-/-}* fibroblasts. The same cell lysates as in **b** were examined for JNK activation by immune-complex kinase assay using GST-c-Jun(1–79) as substrate¹⁵. Substrate phosphorylation was quantified by

phosphorimager, and fold stimulation is indicated. JNK content was analysed by immunoblotting with anti-JNK antibody (Pharmingen). **d**, Transfection of *Ikk β ^{-/-}* fibroblasts with an IKK β expression vector restores transient JNK activation by TNF- α . *Ikk β ^{-/-}* cells were cotransfected with expression vectors encoding Flag-tagged(M2)-IKK β (2 μ g) and HA-JNK1 (1 μ g). After 36 h, cells were treated with TNF- α (20 ng ml⁻¹) as indicated. The activity of HA-JNK1 and expression of HA-JNK1 and M2-IKK β was determined. **e**, **f**, Normal activation of p38 or ERK by TNF- α in *Ikk β ^{-/-}* fibroblasts. The same cell lysates as in **b** were analysed by immunoblotting for activation of p38 (**e**) or ERK (**f**) using corresponding anti-phospho antibodies or pan antibodies (New England Biolabs).

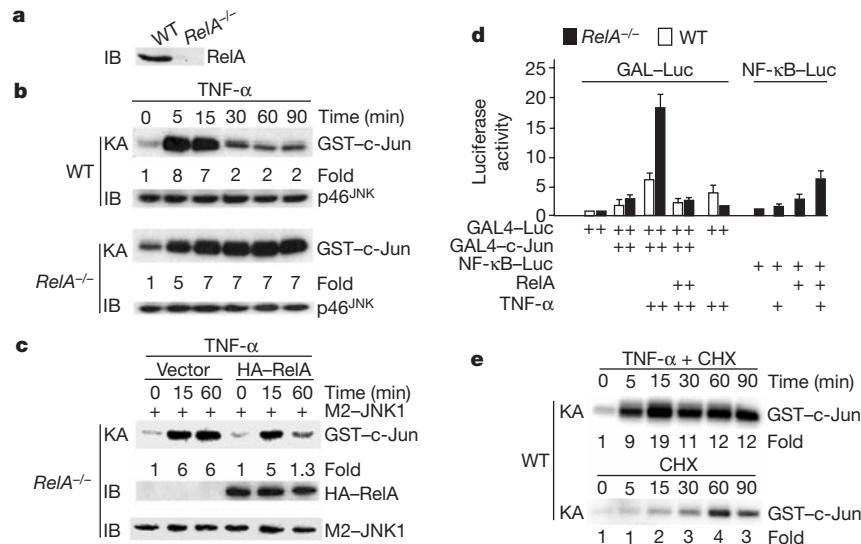


Figure 2 NF-κB mediates the inhibitory effect of IKK on JNK. **a**, Immunoblot analysis of RelA expression in WT and *RelA*^{-/-} fibroblasts with anti-RelA antibody (Zymed). **b**, Activation of JNK by TNF-α is sustained in *RelA*^{-/-} fibroblasts. WT and *RelA*^{-/-} fibroblasts were treated with TNF-α (20 ng ml⁻¹) as indicated, and collected. JNK activity and expression were examined. **c**, Transfection of *RelA*^{-/-} fibroblasts with a RelA expression vector restores transient JNK activation by TNF-α. *RelA*^{-/-} cells were cotransfected with expression vectors encoding M2-JNK1 (1 μg) and either HA-RelA (1 μg) or empty vector (1 μg). After 36 h, cells were treated with TNF-α (20 ng ml⁻¹), as indicated. The activity and expression of M2-JNK1 and expression of HA-RelA were

determined. **d**, Expression of RelA inhibits c-Jun transcriptional activity. WT and *RelA*^{-/-} fibroblasts were cotransfected with a 5× GAL4-Luc reporter plasmid (1 μg), a 2× NF-κB-Luc reporter plasmid (0.25 μg) and expression vectors encoding GAL4-c-Jun(1-223) (2 ng) and RelA (5 ng), as indicated. After 20 h, cells were treated with TNF-α (20 ng ml⁻¹) for 10 h. Cells were collected and relative luciferase activity was determined^{15,21}. The results are presented as means ± standard errors and represent four independent experiments done in duplicate. **e**, TNF-α induces prolonged JNK activation in the presence of CHX. WT fibroblasts were treated with TNF-α (20 ng ml⁻¹) and CHX (10 μg ml⁻¹) or CHX alone. JNK activity was examined.

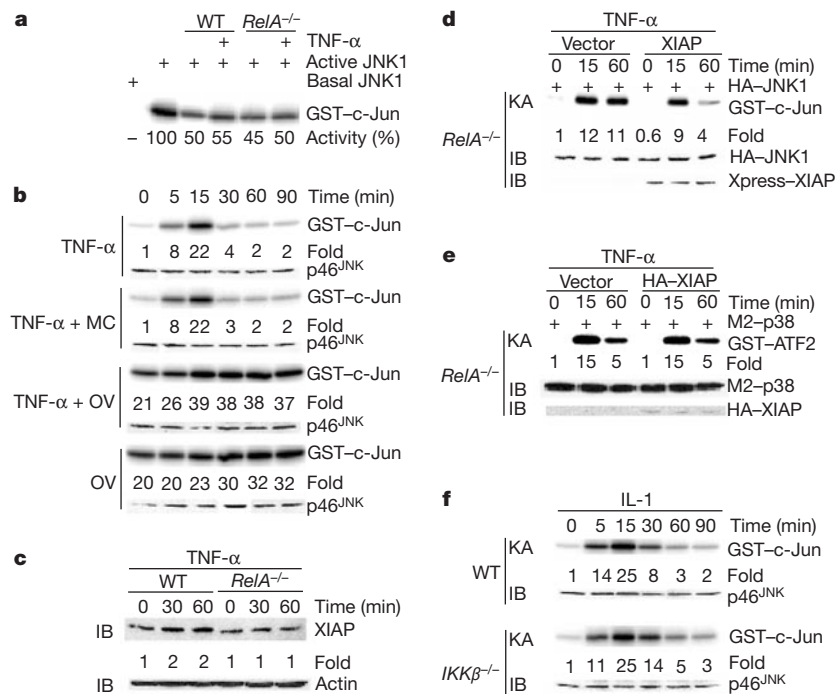


Figure 3 Inhibition of JNK activation by NF-κB is specific to TNF-α signalling and involves XIAP, but not phosphatases. **a**, JNK isolated by immunoprecipitation from non-stimulated (basal) or ultraviolet-irradiated (active) HeLa cells was incubated with untreated or TNF-α-treated WT or *RelA*^{-/-} fibroblast extracts for 1 h in phosphatase buffer²⁴. After extensive washing, JNK activity was examined. The activity of active JNK incubated with buffer alone was given an arbitrary value of 100%. **b**, WT fibroblasts were treated with or without microcystin (MC, 200 nM) or vanadate (OV, 500 μM) for 1 h before treatment with or without TNF-α (20 ng ml⁻¹). JNK activity and expression were examined. JNK activity was expressed relative to the JNK activity of untreated cells, which was given an arbitrary value of 1. **c**, WT and *RelA*^{-/-} cells were treated with TNF-α (20 ng ml⁻¹) as indicated and

expression of XIAP was analysed by immunoblotting using anti-XIAP antibody (PharMingen). The same blot was analysed for expression of actin. **d**, *RelA*^{-/-} cells were cotransfected with expression vectors for HA-JNK1 (2 μg) and either Xpress-tagged XIAP (Xpress-XIAP) (0.5 μg) or empty vector (0.5 μg). After 40 h, cells were treated with or without TNF-α (20 ng ml⁻¹) and the activity and expression of HA-JNK1 and Xpress-XIAP were determined. **e**, *RelA*^{-/-} cells were transiently transfected with M2-p38 (2 μg) and either HA-XIAP (0.5 μg) or empty vector (0.5 μg). After 40 h, cells were treated with or without TNF-α (20 ng ml⁻¹) and the activity and expression of M2-p38 and HA-XIAP were determined. **f**, WT and *IKKβ*^{-/-} fibroblasts were treated with IL-1 (2 ng ml⁻¹) as indicated. JNK activity and expression was determined as described in Fig. 1c.

activation by TNF- α in the absence or presence of phosphatase inhibitors in WT fibroblasts. TNF- α alone induced a transient and robust activation of JNK (Fig. 3b). The serine/threonine phosphatase inhibitor microcystin did not affect TNF- α -induced JNK activation (Fig. 3b). In the presence of the tyrosine phosphatase inhibitor vanadate, TNF- α induced prolonged JNK activation with high basal activity (Fig. 3b). However, vanadate alone elicited prolonged JNK activation (Fig. 3b). The activity of JNK stimulated by TNF- α plus vanadate was similar to the combined JNK activity induced by TNF- α alone and vanadate alone (see Supplementary Information Fig. 1). Thus, induction of phosphatase genes is unlikely to account for the negative crosstalk between the IKK/NF- κ B and JNK signalling pathways. This is consistent with a recent report which shows that inhibition of NF- κ B activation by the super-repressor I κ B α (A32/36) mutant elicits sustained JNK activation without inhibiting mitogen-activated protein kinase phosphatase 1 (MKP1), a JNK phosphatase²⁵.

These results led us to search for other candidates that could function as NF- κ B-induced inhibitors of JNK activation. We tested several known NF- κ B target proteins, including XIAP, c-IAP1, c-FLIP_L, Bcl-X_L and Bcl-2, and found that only transient expression of XIAP inhibited JNK activation by TNF- α in HeLa cells (see Supplementary Information Fig. 2). Furthermore, TNF- α induced XIAP expression in WT, but not *RelA*^{-/-} fibroblasts (Fig. 3c), confirming that the XIAP gene is targeted by NF- κ B⁹. Transfection of *RelA*^{-/-} fibroblasts with a XIAP expression vector restored transient JNK activation in response to TNF- α and modestly decreased TNF- α -stimulated JNK activity (Fig. 3d). These results are consistent with the observation that in *RelA*^{-/-} cells, TNF- α -induced JNK activation was only modestly enhanced whereas the JNK inactivation that occurs after 30 min of TNF- α treatment was profoundly blocked (Figs 1c and 2b). This suggests that XIAP may attenuate TNF- α -mediated JNK activation. Although it has been reported that overexpression of XIAP activates JNK through an unknown mechanism²⁶, we found that overexpression of XIAP simply enhanced expression levels of cotransfected JNK, resulting in higher basal JNK activity. Expression of small amounts of XIAP,

as in our study, suppressed JNK activation by TNF- α without affecting JNK expression levels (Fig. 3d). Ectopic XIAP expression did not affect TNF- α -mediated p38 activation (Fig. 3e), further supporting the specific role of XIAP in NF- κ B-mediated inhibition of JNK activation by TNF- α .

IL-1 and TNF- α often elicit similar biological effects, and their downstream signalling pathways have a great deal of resemblance and overlap. Most importantly, both cytokines are efficient activators of IKK/NF- κ B and JNK signalling pathways²⁷. We therefore examined whether NF- κ B also inhibits JNK activation by IL-1. Surprisingly, IL-1-mediated JNK activation remained transient in *Ikk β* ^{-/-} cells (Fig. 3f). This suggests that inhibition of JNK activation by NF- κ B-induced proteins may be unique to TNF- α .

Sustained JNK activation is suggested to contribute to apoptotic cell death^{23,28}. The prevention of sustained JNK activation by NF- κ B may contribute to its anti-apoptotic activity. To examine this possibility, we used the dominant negative JNKK2(K149M) mutant²⁹ and the constitutively active JNKK2–JNK1 fusion protein³⁰. Expression of JNKK2(K149M) significantly attenuated JNK activation by TNF- α , whereas the JNKK2–JNK1 fusion protein had kinase activity similar to that of TNF- α -activated JNK (Fig. 4a). TNF- α induced apoptosis in *RelA*^{-/-} fibroblasts (Fig. 4b and c; 30% cell death). Expression of JNKK2(K149M) suppressed TNF- α -induced cell death (Fig. 4b and c, 18% cell death), as did expression of the dominant negative JNKK1(257/261AA) mutant (16% cell death) (data not shown). Conversely, TNF- α -induced apoptosis was doubled in *RelA*^{-/-} fibroblasts expressing the JNKK2–JNK1 fusion protein (Fig. 4b and c, 60% cell death). The augmentation of TNF- α -induced cell killing was probably due to the sustained JNK activity of the JNKK2–JNK1 fusion protein, because expression of a kinase-defective JNKK2(K149M)–JNK1 fusion protein had no effect (data not shown). The JNKK2–JNK1 fusion protein alone had no detectable apoptotic effect in *RelA*^{-/-} fibroblasts (data not shown).

Our results show that IKK negatively modulates JNK activity, most probably through the induction of NF- κ B target genes encoding proteins such as XIAP, which interfere with TNF- α -

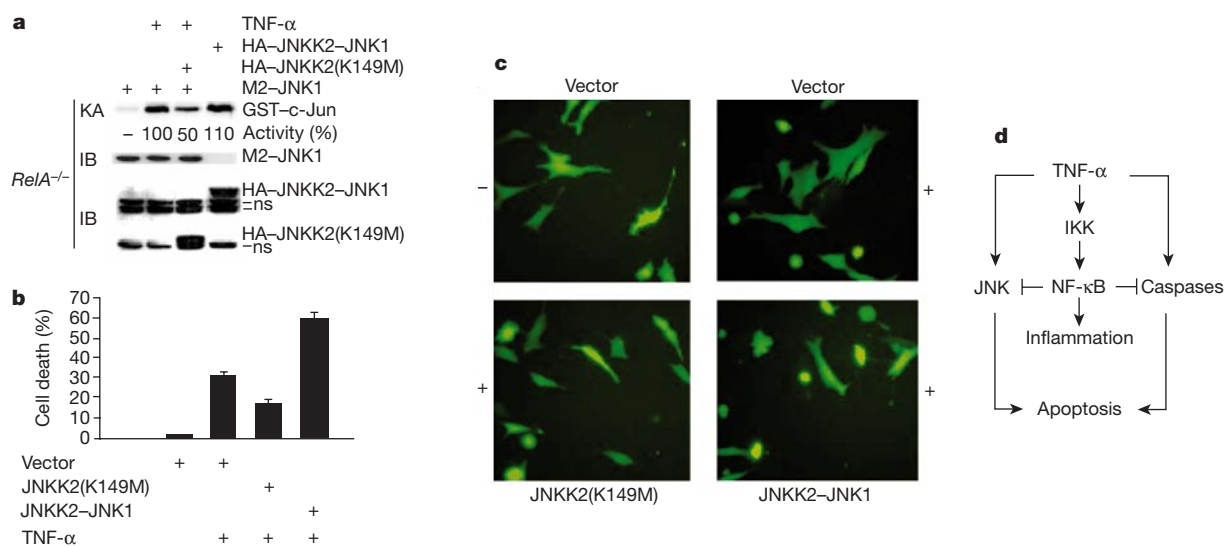


Figure 4 TNF- α killing of *RelA*^{-/-} fibroblasts is inhibited by a dominant negative JNKK mutant but enhanced by a constitutively active JNKK2–JNK1 fusion protein. **a**, *RelA*^{-/-} cells were cotransfected with expression vectors encoding M2–JNK1 (0.7 μ g) and either the JNKK2–JNK1 fusion protein, the dominant negative JNKK2(K149M) mutant or empty vector (2.8 μ g each). Cells were treated with or without TNF- α (20 ng ml⁻¹) for 15 min. The activity of M2–JNK1 and expression of transfected constructs were determined as described^{15,29,30}. **b**, **c**, *RelA*^{-/-} cells were cotransfected with expression vectors encoding GFP (0.7 μ g) and either the HA–JNKK2–JNK1 fusion protein, the dominant negative

HA–JNKK2(K149M) mutant or empty vector (2.8 μ g each). Cells were treated with or without TNF- α (20 ng ml⁻¹) for 6 h, stained with Hoechst dye and visualized by fluorescence microscopy. The death rate of transfected (GFP positive) cells was calculated by counting 12 randomly selected areas (total 40–50 cells per area). Results are presented as means $\pm$ standard errors and represent two individual experiments. The differences in the cell death were statistically significant ($P < 0.05$, Student's *t*-test). +, TNF- α treated, –, untreated. **d**, Schematic illustration of negative regulation of JNK by IKK. See text for details.

mediated, but not IL-1-mediated, JNK activation. This highly specific form of crosstalk allows NF- κ B activation to suppress TNF- α -induced apoptosis through inhibition of both caspases and sustained JNK activation (Fig. 4d). Because apoptosis is an anti-inflammatory process¹ and sustained JNK activation promotes apoptosis, suppression of sustained JNK activation by NF- κ B might enhance TNF- α -induced inflammation (Fig. 4d). □

Methods

Generation of immortalized mouse fibroblasts

WT, *Ikk α ^{-/-}* and *Ikk β ^{-/-}* mouse fibroblasts have been described previously^{19,20}. Cells were immortalized by the 3T3 protocol.

In vitro phosphatase assays

WT or *RelA^{-/-}* cells were treated with or without TNF- α (20 ng ml⁻¹, 60 min). Preparation of cell extracts and phosphatase assays were described previously²⁴.

Apoptosis assays

RelA^{-/-} fibroblasts were cotransfected with expression vectors encoding green fluorescent protein (GFP) and either haemagglutinin-tagged JNKK2-JNK1 (HA-JNKK2-JNK1), HA-JNKK2(K148M) or empty vector at a ratio of 1:4. Under these conditions, cells expressing GFP also expressed either HA-JNKK2-JNK1 or HA-JNKK2(K148M). Cells were treated with or without TNF- α for 6 h and apoptosis was monitored by Hoechst staining for nuclear condensation.

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The 7SK small nuclear RNA inhibits the CDK9/cyclin T1 kinase to control transcription

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The human positive transcription elongation factor P-TEFb, consisting of a CDK9/cyclin T1 heterodimer, functions as both a general and an HIV-1 Tat-specific transcription factor^{1,2}. P-TEFb activates transcription by phosphorylating RNA polymerase (Pol) II, leading to the formation of processive elongation complexes. As a Tat cofactor, P-TEFb stimulates HIV-1 transcription by interacting with Tat and the transactivating responsive (TAR) RNA structure located at the 5' end of the nascent viral transcript³. Here we identified 7SK, an abundant and evolutionarily conserved small nuclear RNA (snRNA) of unknown function^{4,5}, as a specific P-TEFb-associated factor. 7SK inhibits general and HIV-1 Tat-specific transcriptional activities of P-TEFb *in vivo* and *in vitro* by inhibiting the kinase activity of CDK9 and preventing recruitment of P-TEFb to the HIV-1 promoter. 7SK is efficiently dissociated from P-TEFb by treatment of cells with ultraviolet irradiation and actinomycin D. As these two agents have been shown to significantly enhance HIV-1 transcription and phosphorylation of Pol II (refs 6–8), our data provide a mechanistic explanation for their stimulatory effects. The 7SK/P-TEFb interaction may serve as a principal control point for the induction of cellular and HIV-1 viral gene expression during stress-related responses. Our studies demonstrate the involvement of an snRNA in controlling the activity of a Cdk-cyclin kinase.

To identify nuclear factors that can interact with and regulate the activity of P-TEFb, Flag-tagged CDK9 and its associated factors